

Two-Week Summary: April 15–29, 2026

Year 3 Semester B — HUJI Physics

Courses covered (priority order): Mechanics of Continua · Intro to Elementary Particles
· Computational Physics · Analytical Electrodynamics

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1 Mechanics of Continua (77606, Prof. Kol) — PRIMARY

Overview of the Two-Week Arc

Over April 15–29 the course developed the complete theory of **linear isotropic elasticity**. The key milestones were:

1. Deriving Hooke's law from symmetry (effective field theory argument).
2. Cataloguing all elastic moduli and their inter-relations.
3. Deriving the **Lamé–Navier equation**: the elasticity analogue of Laplace's equation.
4. Solving two explicit BVPs: a hanging rod and a *rotating cylinder*.
5. Adding time dependence to obtain the **elastic wave equation** and computing the speeds of longitudinal and transverse sound.

1.1 Kinematics: Strain Tensor

Displacement Field and Linearised Strain

The fundamental kinematic object is the **displacement field**

$$\mathbf{u}(\mathbf{x}) = u_i(\mathbf{x}) \hat{e}_i, \quad (1)$$

giving the vector displacement of the material point that, in the undeformed configuration, lives at $\mathbf{x}$. For *small* displacements the linearised **strain tensor** is

$$\boxed{\varepsilon_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j)}. \quad (2)$$

It is symmetric by construction: $\varepsilon_{ij} = \varepsilon_{ji}$. The antisymmetric part of $\partial_j u_i$ describes infinitesimal rigid rotation and carries no physical deformation.

Physical interpretation of each component:

- *Diagonal elements* ε_{ii} (no sum): fractional elongation along direction i .
- *Off-diagonal elements* ε_{ij} , $i \neq j$: half the angle of shear between the i and j directions.
- *Trace*: $\varepsilon_{kk} = \partial_k u_k = \nabla \cdot \mathbf{u} \approx \Delta V/V$, the fractional volume change.

Hoop Strain in Cylindrical Coordinates

For a **cylindrically symmetric** radial displacement $\mathbf{u} = u_r(r)\hat{e}_r$ the two non-trivial diagonal strains are

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad (3)$$

$$\varepsilon_{\varphi\varphi} = \frac{u_r}{r}. \quad (4)$$